The Energy Cost of Training

Supplementary reading, lesson 9 — Technologies for Artificial Intelligence

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> Not examinable. The habit in Section 6 below — refusing to repeat a carbon figure without knowing which assumptions were stacked to produce it — is not mathematics, but it is what separates a professional from someone quoting a headline.

Why a computer scientist should read this

Section 3.4 of the handout made the network wider and watched what happened. On the acceptance test, the mean of five runs went from **0.6048** with one hidden unit to **0.9445** with eight, within three points of the rig's ceiling of 0.97, and to 0.9549 with sixteen. On the drift problem, two hidden units *can* solve it — Section 3.2 writes such a network down by hand — yet training found that solution in only 6 runs of 20, while eight units found it in 20 of 20. The handout drew the obvious conclusion: a layer wider than the problem strictly needs gives training more chances to start near a good set of lines, "which is why networks used in practice are wider than their task requires."

That conclusion is correct, and at the scale the field now works at it is also an instruction to spend electricity. Making the network bigger is the cheapest lever in Lesson 9 — cheap because a laptop trains a network on 2,250 sensors in seconds. This document is about what the same lever costs when it is pulled industrially, and what a professional is now expected to measure and report.

Three more hooks, all physical rather than statistical. Section 4.4 of the handout noted that one backward sweep costs about as much arithmetic as one forward pass, so a gradient for every weight

costs roughly one more prediction. Arithmetic, and the memory traffic that feeds it, are the two things a chip turns into heat, so Section 4.4 is a claim about watts dressed as a claim about complexity. Section 6.3 says the same thing outright about early stopping: the remedies could not be ranked on accuracy, but "what early stopping certainly bought is time, 39 epochs instead of 200" — and time on a chip is energy. And the drift table of Section 3.4 is a lesson in reliability as much as in accuracy: with two hidden units most runs stopped at three clouds of four, with eight none did. Every restart a more reliable configuration makes unnecessary is a run that never draws power. **Reliability is an energy strategy.**

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1. A price list, starting from this lesson

Section 3.4 of the handout trained twenty networks at each of three widths on the drift problem: sixty networks, for one table of three rows. It then trained five at each of six widths on the acceptance test, and Sections 5.5 and 6.3 ran three per row for two more tables of three rows. That is over a hundred trained networks behind four small tables — the normal ratio. **The published number is the surviving fraction of what was run**, so any energy accounting that counts only the final run understates a project by one or two orders of magnitude. All of it still finishes in minutes on a laptop central processing unit (CPU), which is exactly why the cost never comes up in class.

One step up. Dodge and colleagues, measuring on a commercial cloud in 2022, trained three sizes of DenseNet — a convolutional image classifier — on the handwritten-digit benchmark MNIST (Modified National Institute of Standards and Technology). Each job took 20 to 25 minutes on one graphics processing unit (GPU) and drew **20 to 38 watt-hours**: a 60-watt lamp for half an hour. The same table records five image classifiers trained on ImageNet, from 1.7 kilowatt-hours (kWh) for the smallest to **237.6 kWh** for the largest, which held four GPUs for nine days. A factor of 140 within one model family, and about ten thousand between the digit classifier and the top of that range.

Now the historical anchor, which Lesson 1's supplementary reading introduced. AlexNet was trained for about 90 passes over 1.2 million images, and its authors report **five to six days on two GTX 580 GPUs**, cards rated at 244 watts each. Two of them near full load for five and a half days is about 64 kWh — a rough ceiling on GPU energy alone, excluding host machine, disks and cooling. At the United States average of 0.429 kilograms of carbon dioxide equivalent (CO2e) per kWh that Patterson and colleagues use, that is roughly 28 kg CO2e, or by the Environmental Protection Agency's (EPA) average passenger-vehicle rate **about a hundred kilometres of driving**. The most consequential training run in the modern history of the field cost about one commute. That is the baseline against which every alarming figure since should be read.

What changed was not the cost of one run but their number and size. An OpenAI analysis in 2018 estimated that compute in the largest published training runs grew more than **300,000 times between AlexNet in 2012 and AlphaGo Zero in 2017** — doubling every 3.4 months against Moore's Law's two years. Hardware did not become 300,000 times more efficient in six years, so most of that ratio is real energy. Neither endpoint is a language model: the growth is a property of the field, not of one application of it.

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2. Where the energy actually goes

A chip spends energy on three things: arithmetic, moving numbers to and from where the arithmetic happens, and the overhead of being programmable at all. The canonical measurements are Mark Horowitz's 2014 plenary at the International Solid-State Circuits Conference (ISSCC), for a 45-nanometre process at 0.9 volts. A 32-bit floating-point addition costs about **0.9 picojoules (pJ)**, a multiplication about **3.7 pJ**, an on-chip cache read about **20 pJ**. An access to off-chip dynamic random-access memory (DRAM) costs **1 to 2 nanojoules** — in Horowitz's words, "a couple of orders-of-magnitude higher than the cost of an internal cache access or functional operation." One instruction on a general-purpose processor carries about **70 pJ** of overhead, against "a few pJ for an operation."

Three consequences follow, and each explains hardware you have used. **Moving a number can cost a thousand times more than computing with it**, which is why fast numerical code is a discipline of data locality, and why matrix multiplication is the friendliest operation in computing: $O(n^3)$ arithmetic on $O(n^2)$ data, so the movement amortises. Forward propagation, in the matrix form of the handout's Section 3.3 — the examples stacked as the rows of X, multiplied by $W^{[1]}$ — is exactly this. **Programmability is expensive**, so a processor issuing one instruction per operation spends most of its energy on bookkeeping; widening to many data lanes amortises it, which is what a GPU does, and casting the multiply-accumulate array into silicon amortises it further, which is what a tensor processing unit (TPU) does. **Fewer bits means less movement**, which is most of why training in 16-bit formats became standard.

Put the network of the handout's Section 6.2 on this price list. Two hidden layers of 512 units on 8×8 digits is **301,066 weights and biases**. A forward pass costs about two floating-point operations per parameter per example. Section 4.4 says a backward sweep costs about as much as a forward pass; counted operation by operation it is up to about twice as much, and taking that upper end makes a full step roughly 58 megaflop for a mini-batch of 32 — the size notebook 03 uses — or about **0.13 millijoules of arithmetic** at Horowitz's 4.6 pJ per multiply-add. But those weights occupy 1.2 megabytes, will not sit in a small cache, and reading them once from DRAM costs about **0.23 millijoules** — and they are read at least twice per step. Arithmetic and movement are the same order of magnitude, and movement is the side that grows. This is an illustration on 2014-era figures, not a measurement; absolute energies are far lower today, but the *ratio* has widened rather than narrowed.

There is a memory side to the same coin, which the lighter lesson leaves out and the advanced version in `Advanced/09_neural_networks_advanced/` treats: the backward sweep needs every layer's weighted sums $Z^{[l]}$ and activations $A^{[l]}$ from the forward pass, so they must be **stored** between the two. For the network above that is 256 kilobytes at a mini-batch of 32: nothing. Scale the layers to an image model and it is gigabytes, which is why practitioners recompute activations rather than store them, trading arithmetic back for memory, and why batch size is set by the accelerator rather than by anything statistical. Section 4.1 of the handout chose mini-batches of 32 for a statistical reason — much cheaper than using all the data, and still pointing roughly the right way on average — and at scale that choice carries a hardware clause: the batch size is often not yours to choose.

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3. The datacentre multiplier, and what it hides

Electricity delivered to a server is not electricity drawn from the grid. Cooling, power conversion, lighting and distribution losses consume more. The standard metric is **power usage effective-**

ness (PUE), defined by the industry consortium The Green Grid in 2007 and standardised as ISO/IEC 30134-2:2016 by the International Organization for Standardization and the International Electrotechnical Commission. It is a ratio: total facility energy over energy delivered to the information technology equipment. A PUE of 1.5 means half as much again is spent on everything that is not computing.

The Uptime Institute's annual survey puts the global average near **1.56**, essentially flat for five consecutive years — half the surveyed facilities are over eleven years old, and retrofitting is expensive. Patterson and colleagues cite the United States national average as 1.58 in 2018 and 1.59 in 2020, and report **1.11** for the Google datacentre where their experiments ran. The same computation on identical hardware therefore reaches the meter having cost about 40% more in a typical enterprise facility than in a purpose-built one.

Here is the part students reliably get wrong, and the wrong answer is reasonable. PUE looks like an efficiency metric, so a lower PUE ought to mean less energy per unit of work. It does not, because it is a **ratio with the computing in the denominator**: replace your servers with ones twice as wasteful and your PUE improves, since cooling is now a smaller fraction of a larger total. PUE says nothing about utilisation — a facility idling at 5% load can post an excellent figure — nothing about the carbon intensity of the electricity, nothing about water, and nothing about emissions embodied in manufacturing the hardware. It measures one overhead well and is silent on the rest. The companion metrics that cover part of the gap — water usage effectiveness (WUE), carbon usage effectiveness (CUE), energy reuse factor (ERF) — are far less reported, which is why the European regulation in Section 8 below names them explicitly.

At sector level the record is genuinely surprising. Masanet and colleagues, in *Science* in 2020, found global datacentre electricity roughly flat from 2010 to 2018 — around 1% of world electricity — while the computing work done there grew about **550%**. Efficiency gains, consolidation and the shift from server rooms to hyperscale facilities absorbed nearly the whole increase, which is the strongest available argument against naive extrapolation. It is not a law, though: the International Energy Agency (IEA) put datacentre consumption at **415 terawatt-hours in 2024, some 1.5% of world electricity**, and projected roughly a doubling by 2030. The gains that absorbed the 2010s were largely one-time structural shifts, and they cannot be repeated.

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4. The same run, a different grid

Energy is not carbon. A kilowatt-hour's emissions depend entirely on what generated it, and the spread between grids exceeds almost any efficiency gain available in software. Ember's 2025 review put the **global average at 473 grams of CO2e per kWh in 2024** and the European Union (EU) average at 213. Within Europe, Sweden's power sector was around 11 grams and France's around 41, both dominated by non-fossil generation; Poland's was the Union's highest, in the region of 650 to 700, and India's national figure is around 670. Cleanest to dirtiest here is **more than fifty to one**.

Dodge and colleagues measured what that means for an identical job. The same workload run in sixteen regions of one commercial cloud provider emitted from about **7 kilograms of CO2e in the best region to about 26 in the worst** — same code, same hardware, same duration, differing only in which grid the building is attached to. Their month-by-month lines are nearly flat, so this is a property of the region rather than the season. Patterson and colleagues report the same effect

from inside an operator: the carbon-free fraction of supply varies roughly five to ten times "even within the same country and the same organization."

Time of day matters too, and differently depending on job length. Dodge and colleagues evaluated a scheduler allowed to delay a start by up to 24 hours to catch a cleaner hour. For the short DenseNet runs on digits this cut emissions by about **26%** on average across regions and months. For jobs longer than a day it achieved almost nothing, for a reason that is arithmetic rather than engineering: a multi-day run already averages over the daily cycle, so no peak is left to avoid. For those, pausing and resuming when the grid is dirty — at the cost of wall-clock time — is what pays. Note their own caveat: the optimisations were computed on historical data, so they bound what a real scheduler working from a forecast could achieve.

The upshot is that **the two largest levers on the carbon cost of a run are where you run it and when**, and neither is a machine learning decision. Both are available to a student with a cloud account and a region dropdown, and both dwarf what any amount of tuning will buy.

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5. Training is usually the smaller half

Almost every circulated figure concerns *training*, because training is a discrete event with a start and an end, which makes it easy to measure and easy to dramatise. A deployed model's total energy is training plus every query it will ever answer:

$$\text{footprint} = (E_{\text{train}} + N_{\text{queries}} \cdot E_{\text{inference}}) \times \text{CO2e per kWh}$$

and for anything in production the second term dominates. NVIDIA has estimated that 80–90% of the machine learning workload is inference; Amazon Web Services has claimed 90% of cloud machine learning demand is. Google published its own accounting: machine learning was 10–15% of the company's total energy in each of three consecutive years, split about **three-fifths inference and two-fifths training**. The estimates differ because the populations do — a cloud provider sees other companies' serving traffic — but every one puts inference ahead.

This inverts several intuitions. An architecture costing twice as much to train but 20% less to serve is, on a 10/90 split, a net reduction. A neural architecture search (NAS) is expensive once and then amortised over every model trained in that domain afterwards; Patterson's analogy is that the search is simulating a better light bulb, training is manufacturing the bulb, and inference is every customer switching it on — so charging the simulation to each bulb is a category error.

The table in Section 5.5 of the handout therefore has an economic reading nobody stated in class. "Fit the linear model first" is a claim about accuracy in the handout — on the digits a hidden layer of 32 was worth 2.8 points, for almost four times as many parameters, 2,410 against 650 — and a claim about lifetime serving cost everywhere else. Likewise the table's last row: widening the hidden layer from 32 units to 64 roughly doubled the parameters, 2,410 to 4,810, and moved validation accuracy from 0.9602 to 0.9611, less than the spread between runs. Whatever that last fraction of a point is worth, it is paid for, forever, at every single prediction.

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6. Be careful with other people's numbers

In 2019 Strubell, Ganesh and McCallum published an influential estimate of the carbon cost of a neural architecture search, arriving at **284 tonnes of CO2e** for the search that produced one particular model. It was widely reported, frequently as though it were the cost of training an ordinary model once.

In 2021 Patterson and colleagues — several of them at the organisation that had actually run the search — recomputed it from internal measurements. The earlier estimate, they concluded, was **18.7 times too high for an average organisation and 88 times off in emissions for an energy-efficient one**. Two errors compounded. The search evaluated candidates on small *proxy tasks* and scaled up only the winners, a detail not obvious from the original paper, so the estimate assumed full-size training throughout. And a search is run roughly once per problem domain, not once per model, so attributing its cost to every subsequent training run — as much of the coverage did — multiplies the error again.

Read this correctly, because the wrong reading is tempting in both directions. It is **not** evidence that the concern was manufactured: Strubell and colleagues raised a question the field was not asking, and the measurement culture that followed exists largely because they did. Both papers' recommendations converge on the same fix — stop estimating other people's runs retroactively, measure your own. Nor is it a story about carelessness; Patterson's paper says the estimate was made "despite considerable effort."

It is a story about **stacked assumptions**. A retrospective carbon estimate multiplies a guess at the hardware, a guess at utilisation, a guess at the number of runs, a PUE and a grid factor. Each is individually defensible and uncertain by a factor of 1.5 or 2, and five in a row is one to two orders of magnitude of uncertainty before anyone has made a mistake. Patterson's paper puts the total spread across model, hardware, facility and location at **100 to 1000 times**, then draws the corollary: "These large factors also make retroactive estimates of energy cost difficult."

The advanced version of this lesson, in `Advanced/09_neural_networks_advanced/`, makes this point with gradient checking, which exists because a wrong gradient does not raise an exception — it trains, badly, and looks like a modelling problem. The lighter lesson keeps the habit in miniature: Section 5.2 of the handout counts the weights by hand, $64 \times 32 + 32$ for the hidden layer and $32 \times 10 + 10$ for the output, and checks the total against the 2,410 Keras prints. A wrong carbon estimate does not raise an exception either. It reads as careful work, propagates into slides and press coverage, and is corrected years later if at all. The defence is identical: recompute from the raw inputs by a route that does not reuse the intermediate values.

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7. What is getting better, and what is not

Three trends push the cost down, which is why Patterson's group predicted training emissions would plateau and then shrink. **Specialised hardware** runs this workload 2 to 5 times more efficiently than off-the-shelf systems, by exactly the mechanism Section 2 above gives. **Better models for the same accuracy** — architecture search, distillation, sparsity, reduced precision — buy equal quality for less arithmetic. **Cleaner, better-run facilities** are roughly 1.4 to 2 times more efficient on PUE alone, with siting multiplying through the factor of five to ten in Section 4 above.

Two trends push back, and both are structural. **Diminishing returns:** Thompson and colleagues, examining a decade of image classification among other benchmarks, argue each increment of accu-

racy has been bought with an exponentially larger increment of compute. Lesson 9 shows the same shape at laboratory scale — in the acceptance-test sweep of the handout's Section 3.4 the mean climbed from 0.6048 at one hidden unit to 0.7392 at two, but only from 0.9445 to 0.9549 when eight units became sixteen. The last fraction of a point is always the expensive one, and industrially "expensive" means a grid connection. **Embodied carbon:** everything above concerns *operational* energy, but Gupta and colleagues argue that for modern mobile and datacentre equipment the larger share of lifetime emissions comes from manufacturing the hardware and building the infrastructure — a share that grows as operational energy gets cleaner. An accelerator that halves training energy but is replaced every three years may not be the improvement the operational figure suggests, and essentially no published training-cost figure includes this.

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8. What you will be required to report

Reporting has moved, in about six years, from an activist proposal to a legal obligation.

In papers. Schwartz, Dodge, Smith and Etzioni's "Green AI" (2020) proposed that efficiency be reported alongside accuracy as a first-class result, with floating-point operations as the hardware-independent measure. Henderson and colleagues built a framework for systematic energy and carbon reporting; Lacoste and colleagues built a calculator turning hardware, hours and region into an estimate; open tooling now logs energy from inside a training script. Major conferences ask about compute in their submission checklists, MLPerf added power measurement, and model cards — the documentation format proposed by Mitchell and colleagues in 2019 — increasingly carry a compute and emissions section.

In law, for buildings. The EU's Energy Efficiency Directive (Directive (EU) 2023/1791), Article 12, with Commission Delegated Regulation (EU) 2024/1364, obliges datacentres with installed information technology power demand of **500 kilowatts or more** to report annually to a European database: total energy consumption, PUE, water usage effectiveness, energy reuse factor and renewable energy factor, across roughly two dozen data points, by 15 May for the previous calendar year. The mechanism is disclosure, not a cap, and its immediate effect is that PUE and its companions stop being voluntary marketing figures and become filed numbers.

In law, for models. The EU Artificial Intelligence Act (Regulation (EU) 2024/1689), Annex XI, requires providers of general-purpose artificial intelligence (AI) models to document the **known or estimated energy consumption** of the model, and says that where it is unknown the estimate may be based on the computational resources used. The limits matter as much as the requirement: it covers energy rather than emissions, excludes embodied carbon, prescribes no methodology, and directs disclosure to regulators rather than the public.

Within the working life of anyone reading this, "how much energy did that cost?" moves from a question nobody asks to a field on a form — and the person who fills it in is the one who ran the job.

What to take from it

- **Scaling is the cheapest lever in Lesson 9 and the most expensive one in industry.** Widening the hidden layer from one unit to eight took the acceptance test from 0.6048 to

0.9445 and cost seconds on a laptop; the same trade at ImageNet scale runs to hundreds of kilowatt-hours, and the published tables are the surviving fraction of what was actually run.

- **Moving a number costs far more than computing with it** — about a thousand times more for a DRAM access than a floating-point operation. That one ratio explains specialised accelerators, reduced precision, the obsession with data locality, and why fetching the weights of the handout's 301,066-parameter network costs more energy than the arithmetic done with them.
- **The two biggest levers on carbon are where and when you run, and neither is a modelling decision.** Grid intensity varies more than fiftyfold between national grids and roughly fourfold across one provider's regions, and delaying a short job by a few hours cut measured emissions by about a quarter. Serving, meanwhile, dominates a deployed model's lifetime — so a slightly worse, much cheaper model is often the better engineering decision.
- **Treat any retrospective carbon figure as the product of five uncertain assumptions.** The most famous one was later found 18.7 to 88 times too high, by people with the actual measurements. Ask which hardware, which utilisation, which PUE, which grid, how many runs — and if the paper does not say, the number is an illustration, not a measurement.

Where to look next

Resource	Type	Why read it
Horowitz, "Computing's Energy Problem (and what we can do about it)", ISSCC 2014	Conference plenary	The price list of Section 2 above, from the person who measured it; it reframes what a fast program is
Masanet et al., "Recalibrating global data center energy-use estimates", *Science* 367 (2020)	Paper	550% more compute for flat energy, 2010–2018 — the best argument against naive extrapolation
Patterson et al., "Carbon Emissions and Large Neural Network Training", arXiv:2104.10350 (2021)	Paper	The PUE and grid factors, the training/inference split, and the corrected NAS estimate with its post-mortem
Dodge et al., "Measuring the Carbon Intensity of AI in Cloud Instances", FAccT 2022	Paper	Real measurements across 16 regions and a year, plus the scheduling results of Section 4 above
Schwartz, Dodge, Smith & Etzioni, "Green AI", *Communications of the ACM* 63(12) (2020)	Position paper	The case for reporting efficiency as a result, and why floating-point operations were proposed as the measure
Gupta et al., "Chasing Carbon: The Elusive Environmental Footprint of Computing", HPCA 2021	Paper	The embodied-carbon half that operational figures leave out
Directive (EU) 2023/1791 Art. 12 and Delegated Regulation (EU) 2024/1364	Primary legal text	The datacentre reporting obligation in full, at `eur-lex.europa.eu`

Resource	Type	Why read it
ISO/IEC 30134-2:2016	Standard	PUE as actually defined, with the measurement categories that make two quoted figures incomparable
IEA, *Energy and AI* (2025)	Official report	The sector-level projections, with their assumptions stated
Hennessy & Patterson, *Computer Architecture: A Quantitative Approach*, ch. 7, "Domain-Specific Architectures"	Book chapter	Why a TPU exists, derived from the energy numbers rather than asserted